System Validation Report — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Team: Kshitij G Shettigar · Kartik Basavaraj Meti · Komal S Sajja
Prepared by: Kartik Basavaraj Meti (v1.0, 2026-11-02) · **Reviewed and re-validated by:** Kshitij G Shettigar (v1.1, 2026-11-03; v1.2, 2026-11-11: Defect 1 fixed and re-tested, §5; v1.3, 2026-11-12: 200-user load test, §4.1)
Build tested: commit `2d74453` (the frozen Sprint 2 build) · **Baseline:** `docs/SRS.pdf` v3.0, `docs/Test_Plan.pdf` v4.0

1. Purpose

This report validates whether the Online Examination Portal at commit `2d74453` satisfies the requirements of `docs/SRS.pdf`. It covers the 13 system test cases (ST_001–ST_013) of `docs/Test_Plan.pdf` and a check of every non-functional requirement (NFR-001 to NFR-014).

What changed in v1.1. Version 1.0 (Kartik, 2 Nov) ran the automated suite and walked five system cases by hand. In review, the repository owner re-ran all 13 system cases and the manual NFR checks through the real user interface in Chrome, Edge and Firefox (§3). That run found one defect, in FR-028, which v1.0 had reported as "no defects" (§5). It also showed that NFR-001 and NFR-003 have not been verified (§4). §6 lists every correction to v1.0.

What changed in v1.3. v1.1 missed PR #87, which ran the full 200-user load test on 28 Oct. The test was run three more times on 12 Nov with a script that also opens question pages. NFR-001 and NFR-003 now pass, with one failing run disclosed (§4.1).

2. Environment

Item	Run 1 — v1.0 (Kartik, 2 Nov)	Run 2 — v1.1 (Kshitij, 3 Nov)
OS	Fedora Linux 44 (x86_64)	Windows 11 Home (build 26300)
Python / Django	Python 3.14.8 / Django 5.2.18	Python 3.10.11 / Django 5.2.18
Database	SQLite 3 (development configuration)	SQLite 3, a throwaway file seeded with the Test Plan's fixtures (OEP_DB_PATH)
Server	`manage.py runserver`	`manage.py runserver` with DEBUG off, as in production
Browsers	Playwright Chromium 156.0.8078.4 · Playwright Firefox 157.0	Google Chrome 156.0.8078.80 · Microsoft Edge 156.0.4314.58 · Firefox 157.0, driven by Playwright 1.64
Automated suite	`manage.py test -v 2`: `Ran 227 tests in 151.475s — OK`	(the same suite, already recorded in `docs/Unit_Test_Report.pdf`)
Accessibility	`@axe-core/playwright`	axe-core 4.10.3

How run 2 works. A script seeds the fixtures through the ORM (the Test Plan's preconditions). Then every step under test goes through the browser: forms are filled, buttons clicked and pages read exactly as a user would. The script checks the database afterwards and takes the screenshots. Run 2 takes about 18 minutes, because ST_007, ST_009, ST_010 and ST_012 wait for real time to pass (a window to open, a 3-minute disconnect, a 3-minute paper, a window to close). The script is kept with the team's working files, not in the repository.

In the first pass, ST_012 stopped on an error in the script, not the application: the NFR-008 erasure check had erased one of the five students ST_012 signs in. The script was changed to erase a different student, and ST_011,

NFR-008 and ST_012 were run again together on a fresh database. The results below for those three are from that second pass.

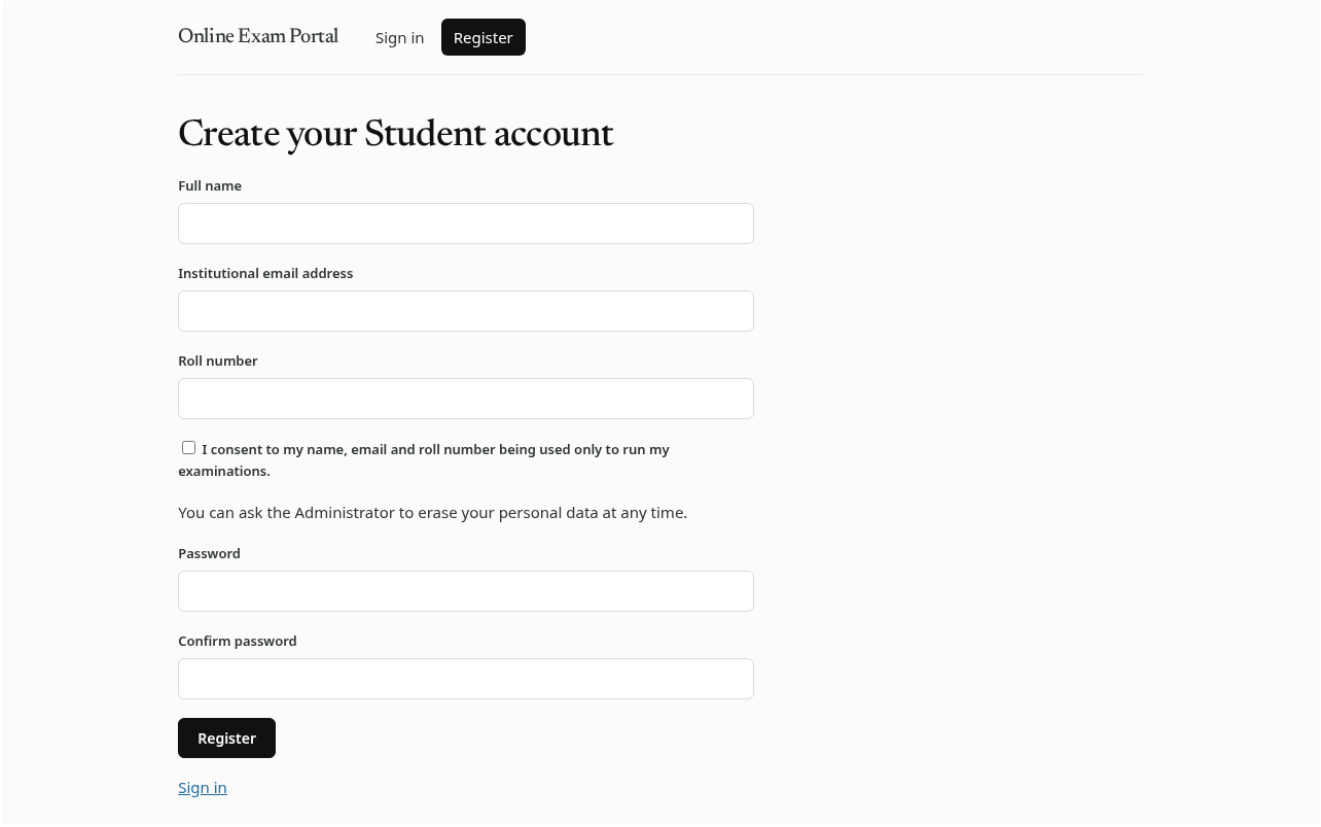
3. System test results

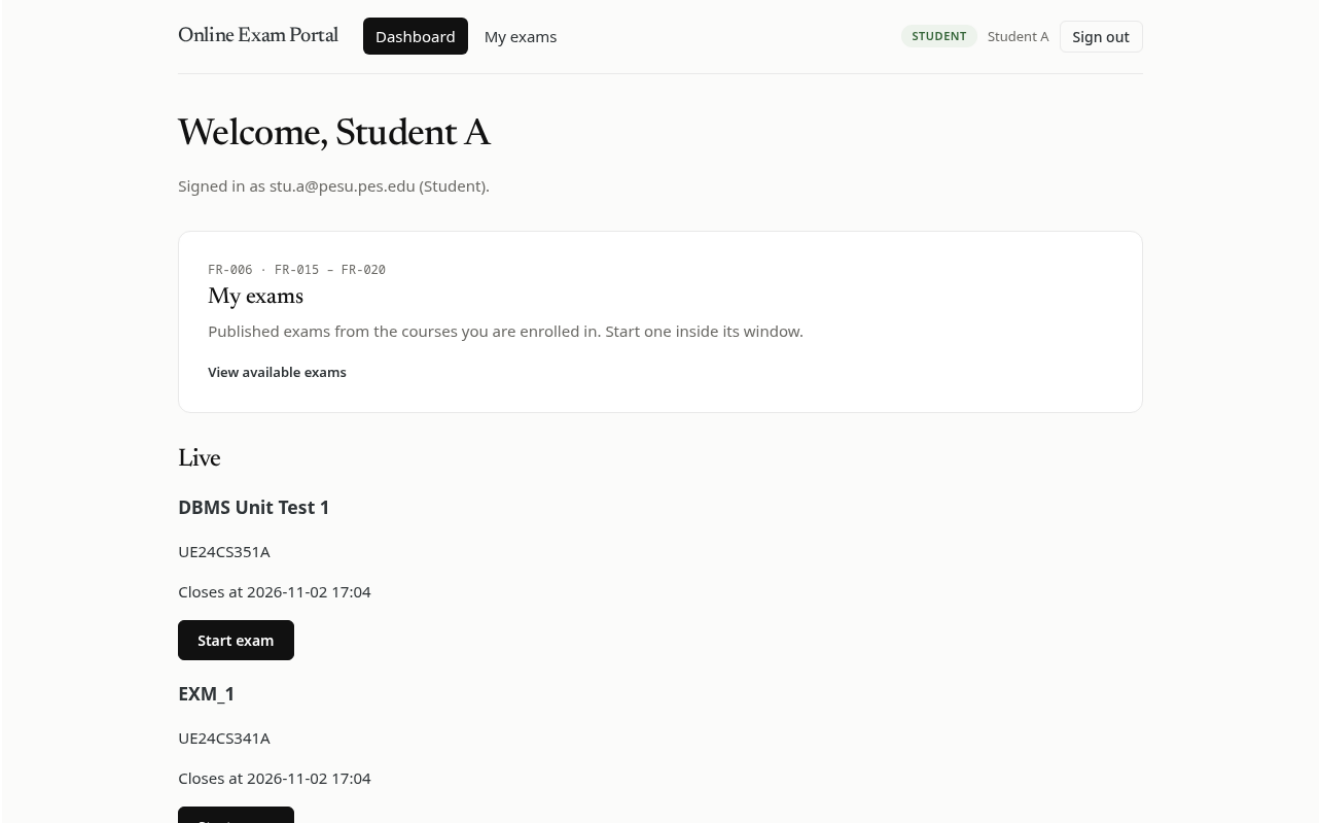
ID	Requirements	Automated test (run 1)	Manual run 1	Browser run 2	What run 2 observed	Result
ST_001	FR-001, FR-002	`test_ST_001_register_sign_in_and_load_student_dashboard` — Pass	Pass	Pass	New account registered through `/register/` with consent, signed in, landed on `/student/`	**Pass**
ST_002	FR-002, FR-003	`test_ST_002_each_role_lands_on_its_own_dashboard` — Pass	Pass	Pass	Each role lands on its own dashboard; Student → `/faculty/questions/` and Faculty → `/admin/courses/` both HTTP 403	**Pass**
ST_003	FR-004	`test_ST_003_admin_creates_and_archives_course` — Pass	—	Pass	Course created and assigned; listed for FAC_A; after archiving it is absent from the exam form's course list, roster still readable	**Pass**
ST_004	FR-005, FR-006	`test_ST_004_end_to_end_cohort_isolation_between_two_courses` — Pass	—	Pass	Each Student sees only their own course's exam; the other course's exam URL is HTTP 403	**Pass**
ST_005	FR-007, FR-008, FR-009	`test_ST_005_author_tag_publish_then_edits_become_versions` — Pass	Pass	Pass	Three questions authored with topics and difficulties; editing one after publication created a new version, and the paper still holds the original	**Pass**
ST_006	FR-010, FR-011, FR-012	`test_ST_006_paper_assembly_and_publication_keep_the_policy_values` — Pass	—	Pass	25 bank questions added; the Draft is not listed for an enrolled Student (direct URL HTTP 404); after Publish it is listed with 60 min, pass mark 40%, negative marking on, total 50	**Pass**
ST_007	FR-013, FR-014	`test_ST_007_exam_moves_upcoming_live_completed_as_server_time_advances` — Pass	Pass	Pass	Faculty scheduled a window 2 minutes ahead: **Upcoming**; when it opened: **Live** with a Start button; after submitting: **Completed**	**Pass**
ST_008	FR-015–FR-018	`test_ST_008_paper_delivery_under_a_live_window` — Pass	—	Pass	Start refused before the window (no attempt created); inside it the timer read 60:00 then 59:57; one question per page; palette for questions 1–5 read answered, answered, marked, not-answered, not-visited	**Pass**
ST_009	FR-019, FR-021	`test_ST_009_answer_change_clear_resume_and_submit` — Pass	—	Pass	Answered 5, changed 2, cleared 1, closed the browser for 3 minutes; on return all five states were restored exactly and the timer had kept running (56:57 after 3.1 min)	**Pass**
ST_010	FR-020	`test_ST_010_three_minute_paper_end`	—	Pass	A 3-minute paper left open: at zero the page showed	**Pass**

ID	Requirements	Automated test (run 1)	Manual run 1	Browser run 2	What run 2 observed	Result
		`_to_end` — Pass			"Time is up. Your paper has been submitted."; submitted at start + 3:00 exactly; 3 answers kept	
ST_011	FR-022, FR-023, FR-024	`test_ST_011_five_students_are_scored_with_no_faculty_action` — Pass	—	Pass	Five students sat EXM_1 through the UI; scores 50, 37.5, 17.5, 11, 0 with no Faculty action; 37.5 (75%, Pass) and 11 (22%, Fail) match the hand calculation	**Pass**
ST_012	FR-025, FR-026, FR-027	`test_ST_012_end_to_end_publication_and_scorecard` — Pass	—	Pass	Before publication the scorecard URL is HTTP 403 with no score; Faculty published after the window closed and each of the five Students' scorecards showed the stored score (50, 37.5, 17.5, 11, 0); after Unpublish the URL is HTTP 403 again and no stored score changed	**Pass**
ST_013	FR-028, FR-029, FR-030	`test_ST_013_admin_reads_the_log_filtered_to_one_exam` — Pass	Pass	**Fail**	No answer key in any of the 7 payloads the browser received; the Administrator's audit log, filtered to the attempt, shows create, submit and evaluate, and filtered to the exam, the results publication. But the attempt carries **6 focus-loss events for 2 tab switches** — Defect 1	**Fail** (Defect 1)

Summary: 12 of 13 system test cases pass on `2d74453`. ST_013 fails on Defect 1 (§5); it passes after the 11 Nov fix (run 3). The automated ST_013 test passes because the Django test client runs no JavaScript, and the defect is in the page's script.

3.1 Screenshots — run 1 (Kartik)





Online Exam Portal Dashboard My exams STUDENT Student A Sign out

Request could not be completed

Please check your request or try again later.

Online Exam Portal Sign in Register

Sign in

Email address

Password

Sign in

Register

Online Exam Portal Dashboard My courses Question bank Examinations FACULTY Faculty A Sign out

Question bank

Create question

COURSE	QUESTION	TOPIC	DIFFICULTY	MARKS	VERSION	ACTIONS
UE24CS341A	Question 1	t	Easy	+2.00 / −0.50	v1	Edit question 1 Delete question 1
UE24CS341A	Question 2	t	Easy	+2.00 / −0.50	v1	Edit question 2 Delete question 2
UE24CS341A	Question 3	t	Easy	+2.00 / −0.50	v1	Edit question 3 Delete question 3
UE24CS341A	Question 4	t	Easy	+2.00 / −0.50	v1	Edit question 4 Delete question 4

Online Exam Portal Dashboard My exams STUDENT Student A Sign out

Welcome, Student A

Signed in as stu.a@pesu.pes.edu (Student).

FR-006 · FR-015 – FR-020

My exams

Published exams from the courses you are enrolled in. Start one inside its window.

View available exams

Live

DBMS Unit Test 1

UE24CS351A

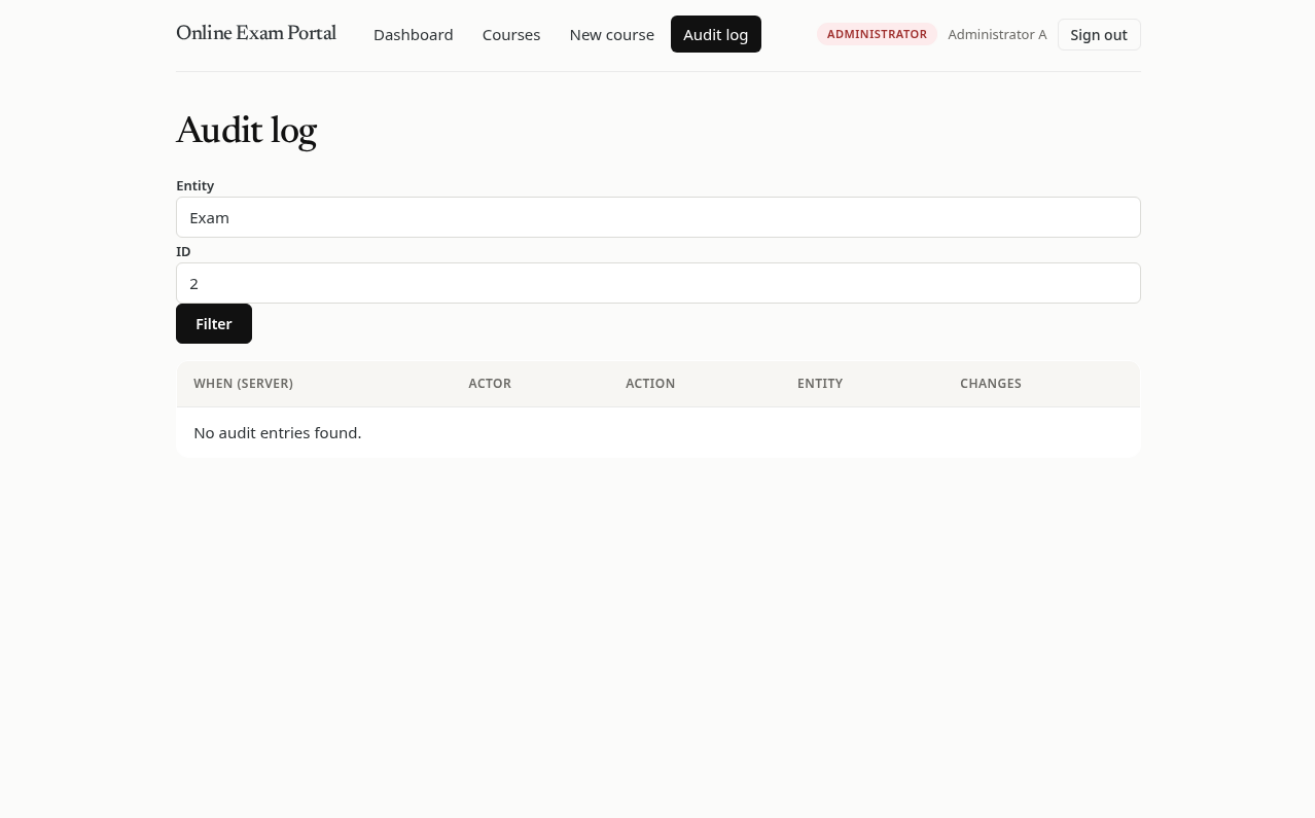
Closes at 2026-11-02 17:04

Start exam

EXM_1

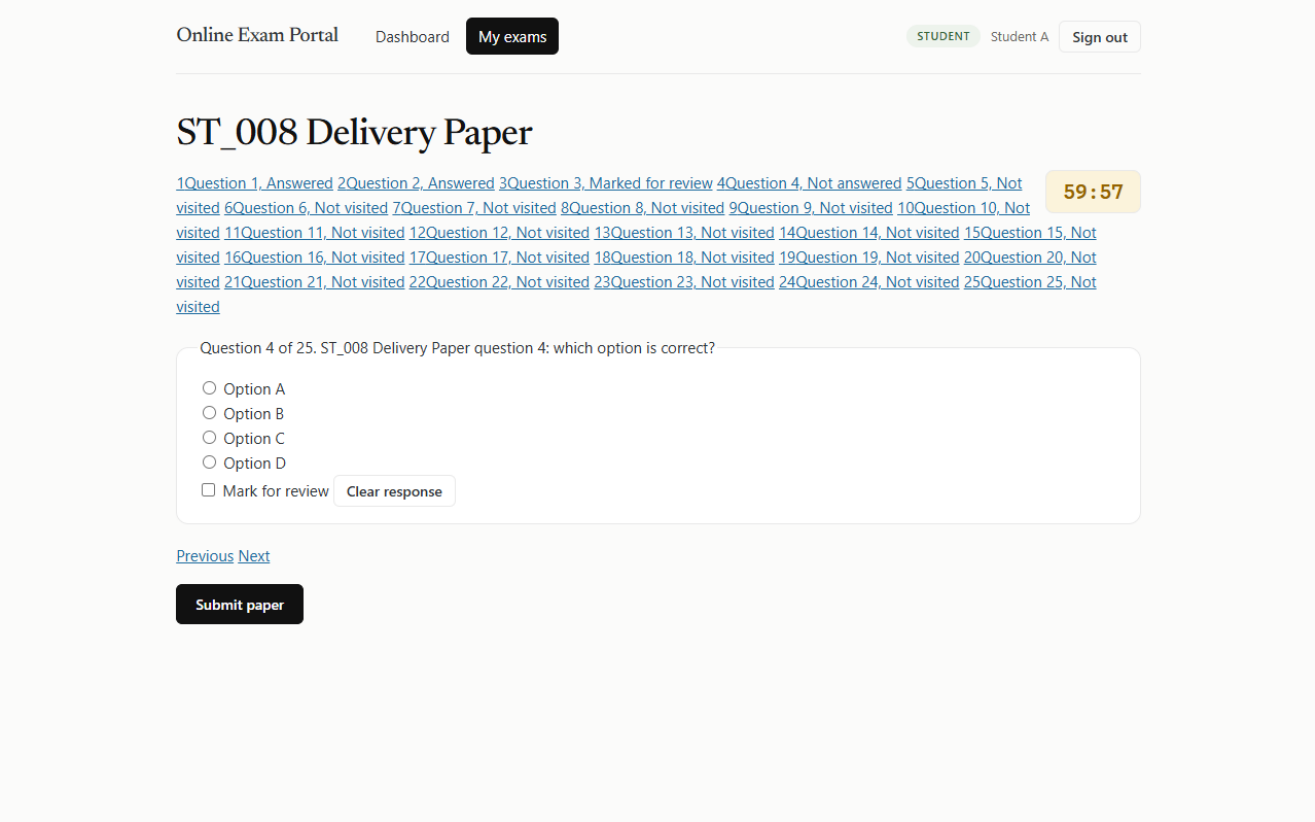
UE24CS341A

Closes at 2026-11-02 17:04



The remaining run 1 screenshots are in `docs/validation/`.

3.2 Screenshots — run 2 (re-validation)



Online Exam Portal Dashboard My exams STUDENT Student A Sign out

ST_009 Resume Paper

1Question 1, Answered 2Question 2, Answered 3Question 3, Answered 4Question 4, Not answered 5Question 5, Answered
6Question 6, Not visited 7Question 7, Not visited 8Question 8, Not visited 9Question 9, Not visited 10Question 10, Not visited
11Question 11, Not visited 12Question 12, Not visited 13Question 13, Not visited 14Question 14, Not visited 15Question 15, Not visited
16Question 16, Not visited 17Question 17, Not visited 18Question 18, Not visited 19Question 19, Not visited 20Question 20, Not visited
21Question 21, Not visited 22Question 22, Not visited 23Question 23, Not visited 24Question 24, Not visited 25Question 25, Not visited

56:57

Question 1 of 25. ST_009 Resume Paper question 1: which option is correct?

- ○ Option A
- ◉ Option B
- ○ Option C
- ○ Option D

☐ Mark for review Clear response

Next

Submit paper

Online Exam Portal Dashboard My exams STUDENT Student A Sign out

ST_010 Three-minute Paper

Time is up. Your paper has been submitted.

Results not yet published

Back to dashboard

EXM_1 DBMS Unit Test 1 · Scorecard

Total: 37.50 / 50.00

Percentage: 75.00%

Outcome: Pass

Marks on each question

#	QUESTION	YOUR ANSWER	MARKS
1	EXM_1 DBMS Unit Test 1 question 1: which option is correct?	Option B	2.00
2	EXM_1 DBMS Unit Test 1 question 2: which option is correct?	Option B	2.00
3	EXM_1 DBMS Unit Test 1 question 3: which option is correct?	Option B	2.00
4	EXM_1 DBMS Unit Test 1 question 4: which option is correct?	Option B	2.00
5	EXM_1 DBMS Unit Test 1 question 5: which option is correct?	Option B	2.00
6	EXM_1 DBMS Unit Test 1 question 6: which option is correct?	Option B	2.00
7	EXM_1 DBMS Unit Test 1 question 7: which option is correct?	Option B	2.00
8	EXM_1 DBMS Unit Test 1 question 8: which option is correct?	Option B	2.00
9	EXM_1 DBMS Unit Test 1 question 9: which option is correct?	Option B	2.00
10	EXM_1 DBMS Unit Test 1 question 10: which option is correct?	Option B	2.00
11	EXM_1 DBMS Unit Test 1 question 11: which option is correct?	Option B	2.00
12	EXM_1 DBMS Unit Test 1 question 12: which option is correct?	Option B	2.00
13	EXM_1 DBMS Unit Test 1 question 13: which option is correct?	Option B	2.00
14	EXM_1 DBMS Unit Test 1 question 14: which option is correct?	Option B	2.00
15	EXM_1 DBMS Unit Test 1 question 15: which option is correct?	Option B	2.00
16	EXM_1 DBMS Unit Test 1 question 16: which option is correct?	Option B	2.00
17	EXM_1 DBMS Unit Test 1 question 17: which option is correct?	Option B	2.00
18	EXM_1 DBMS Unit Test 1 question 18: which option is correct?	Option B	2.00
19	EXM_1 DBMS Unit Test 1 question 19: which option is correct?	Option B	2.00
20	EXM_1 DBMS Unit Test 1 question 20: which option is correct?	Option B	2.00
21	EXM_1 DBMS Unit Test 1 question 21: which option is correct?	Option A	-0.50
22	EXM_1 DBMS Unit Test 1 question 22: which option is correct?	Option A	-0.50
23	EXM_1 DBMS Unit Test 1 question 23: which option is correct?	Option A	-0.50
24	EXM_1 DBMS Unit Test 1 question 24: which option is correct?	Option A	-0.50
25	EXM_1 DBMS Unit Test 1 question 25: which option is correct?	Option A	-0.50

Back to dashboard

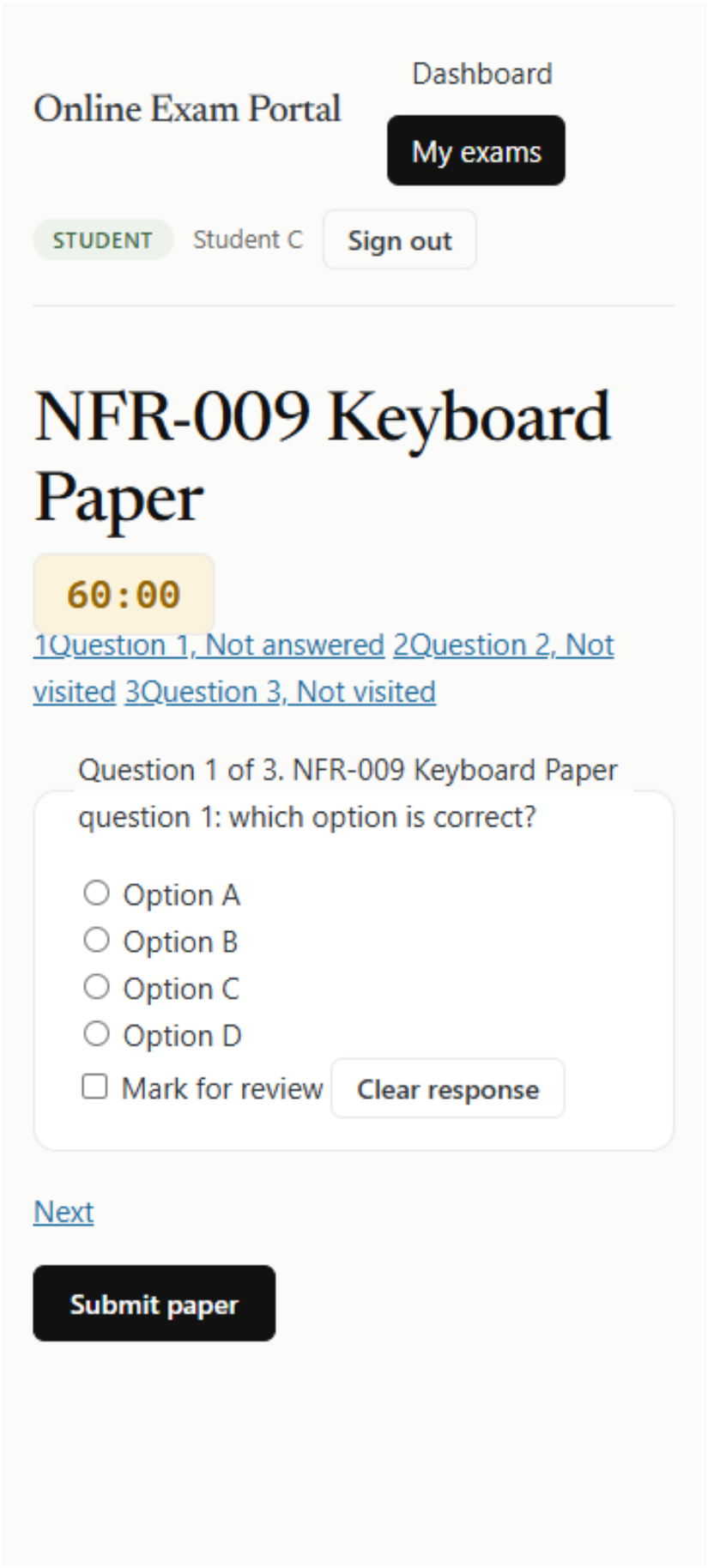
Online Exam Portal Dashboard My courses Question bank Examinations FACULTY Faculty A Sign out

FR-028 Focus Paper · Integrity events

Each row is the examination window losing focus, timed by the server. These are evidence for review, not a verdict.

STUDENT	ROLL NUMBER	FOCUS LOSSES	SERVER TIMES
Student C	PES1UG24CS003	7	2026-11-03 19:27:47, 2026-11-03 19:27:48, 2026-11-03 19:27:48, 2026-11-03 19:27:49, 2026-11-03 19:27:49, 2026-11-03 19:27:50, 2026-11-03 19:27:50

Back to the examination



Every run 2 screenshot is in `docs/validation/rerun/`.

4. Non-functional requirements

NFR	SRS pass criterion	How validated	Result	Evidence
NFR-001	"a Locust load test with 200 simulated students shows a 95th-percentile response time ≤ 2 s"	Four 200-user Locust runs against Gunicorn: L1 (PR #87, 28 Oct) and L2–L4 (12 Nov)	**Pass** — p95 1.5–1.9 s in L1, L3, L4; L2 3.6 s, disclosed (§4.1)	§4.1; `loadtest/locustfile.py`
NFR-002	"a timed test of evaluating a 100-question attempt completes in ≤ 5 s"	`test_IT_030_NFR_002_100_question_attempt_is_evaluated_within_5_seconds`	Pass	test run
NFR-003	"NFR-001's load test passes with the bank seeded to 50,000 questions"	Every NFR-001 run had `seed_bank`'s 50,000 questions loaded	**Pass**, with NFR-001	§4.1
NFR-004	"killing and restarting the server during an attempt restores service in ≤ 60 s and the attempt resumes intact"	`test_NFR_004_service_returns_within_60_s_and_the_attempt_resumes_intact`	Pass	test run
NFR-005	"the attempt flow completes on each browser and at 360 px width without horizontal scrolling"	Run 1: Chromium and Firefox. Run 2: a whole attempt started, answered and submitted (score 6.00) in Chrome 156, Edge 156 and Firefox 157; at 360 px the dashboard, attempt page and scorecard are each 360 px wide	Pass	`NFR_005_*.png`, `rerun/NFR-005_*.png`
NFR-006	"the GitHub history shows no direct commits to main after 19 Oct, and every merged PR has a reviewer other than its author"	GitHub history and `docs/Deviations.pdf`	**Pass with recorded exceptions** — admin merges without a second reviewer are logged in D-04, D-08, D-10, D-11, D-12	`docs/Deviations.pdf`
NFR-007	"the coverage table of docs/Test_Plan.pdf lists all 30 FRs"	The four `test_NFR_007_*` tests in `portal.test_traceability`	Pass	test run
NFR-008	"registration cannot complete without ticking consent; an Administrator can erase a Student's personal data"	Run 1: the browser's `required` attribute blocks the form. Run 2 removed that attribute and submitted anyway: the **server** refused and no account was created. An Administrator then erased a Student: sign-in with the old credentials is refused and the stored name and email are replaced	Pass	`NFR_008_*.png`, `rerun/NFR-008_*.png`
NFR-009	"an axe/Lighthouse audit reports no AA violations, and the whole attempt can be completed with the keyboard alone"	Run 1: 0 violations on login, dashboard and attempt pages. Run 2: axe-core 4.10.3 (WCAG 2.1 A/AA) on the attempt page: 0 violations; a 3-question attempt answered and submitted with Tab, arrow keys and Enter only (Submit reached in 13 Tabs), score 6.00	Pass	`NFR_009_axe.png`, `rerun/NFR-009_keyboard.png`
NFR-010	"code review finds no hard-coded pass mark or negative-marking value; changing them on an examination changes its scoring"	The six `test_NFR_010_*` tests in `evaluation.test_policy`	Pass	test run
NFR-011	"the password column holds only pbkdf2_sha256$… hashes, and the logs contain no password after a registration and a sign-in"	`test_NFR_011_passwords_are_hashed_and_not_logged`; run 2 read the password column: all 11 usable passwords start with `pbkdf2_sha256$1000000$`, and the erased account holds Django's unusable-password marker	Pass	test run, run 2 query

NFR	SRS pass criterion	How validated	Result	Evidence
NFR-012	"the 6th sign-in attempt after 5 failures is refused even with the right password, and succeeds after 15 minutes"	`test_IT_038_sixth_attempt_with_right_password_is_refused`, `test_IT_038_lock_lifts_after_15_minutes`	Pass	test run
NFR-013	"a POST without a CSRF token is refused with HTTP 403; a code search finds no raw SQL; a forced error shows a generic page with DEBUG=False"	Run 2: POST `/login/` without a token → HTTP 403; `/nope/` → HTTP 404 generic page with no stack trace; `git grep` finds no `.raw(`, `cursor(` or `.extra(` outside tests and migrations	Pass	`NFR_013_error.png`, `rerun/NFR-013_error.png`
NFR-014	"a secret scan of the public repository finds no secret, and the app refuses to start without the SECRET_KEY variable"	`manage.py check` without `DJANGO_SECRET_KEY` exits 1 with `KeyError: 'DJANGO_SECRET_KEY'`; a scan of all 200 commits of history for key literals, `django-insecure-` keys, AWS and GitHub tokens finds only the README's `dev-only` placeholder	Pass	run 2 output

Summary: 13 of 14 pass, and NFR-006 passes with recorded exceptions. (v1.1: 11 of 14, with NFR-001 and NFR-003 not verified.)

4.1 Load test (NFR-001, NFR-003)

Setup. Gunicorn with 25 workers in WSL 2 Ubuntu 22.04 on a 12-thread laptop, Locust 2.46 on the same machine. Each run started from a fresh copy of a SQLite file seeded with 50,000 questions and 200 enrolled students.

Load. 200 students added at 20 per second. Each signs in, starts the exam, opens questions, answers them and polls the timer, then submits. `--reset-stats` starts the count once all 200 are in, so the figures are "while 200 attempts are active".

Script change on 12 Nov. L1's script answered the first question over and over and never opened another question page. L2–L4 open a random question from the palette before each answer, which adds FR-018's page load, the main page a student requests.

Run	Date	Script	Length	Requests	Failures	Aggregate p95	Result
L1	28 Oct (PR #87)	Answer and timer only	5 min	26,461	0	1.9 s	Pass
L2	12 Nov	With question pages	5 min	32,336	2 (HTTP 500 on `answer`)	**3.6 s**	**Fail**
L3	12 Nov	With question pages	3 min	23,972	0	1.8 s	Pass
L4	12 Nov	With question pages	5 min	43,439	0	1.5 s	Pass

Verdict: Pass. Three of four runs meet the 2 s target with no failed request, including two with question pages. L2 is reported rather than dropped.

Why L2 failed. The CPU was 55–65% idle, so the server was not short of compute. Throughput dropped in bursts while the CPU sat idle, which is waiting on a lock rather than work.

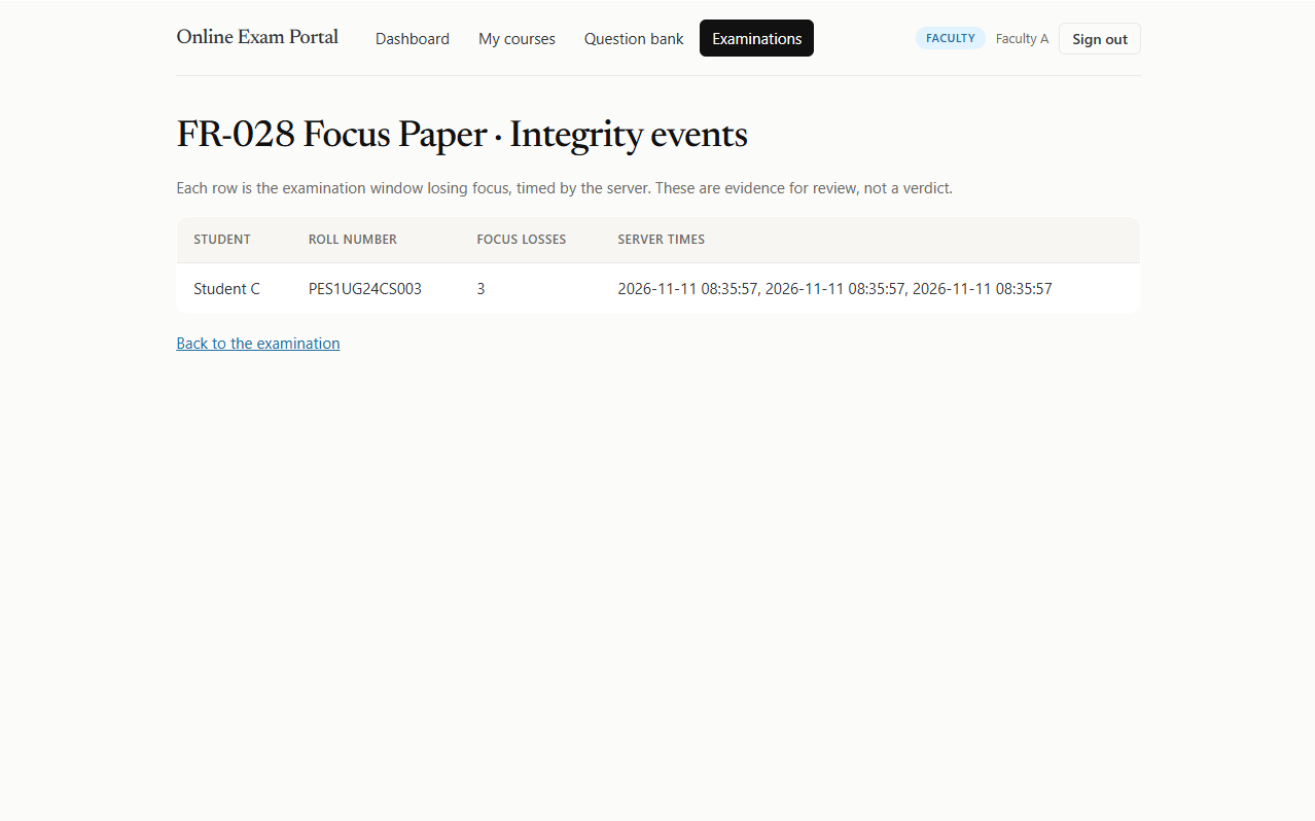
The two 500s came after 21.8 s, just past SQLite's 20 s busy timeout, so they were almost certainly "database is locked". Every answer save writes, about 50 a second here, and SQLite has one writer.

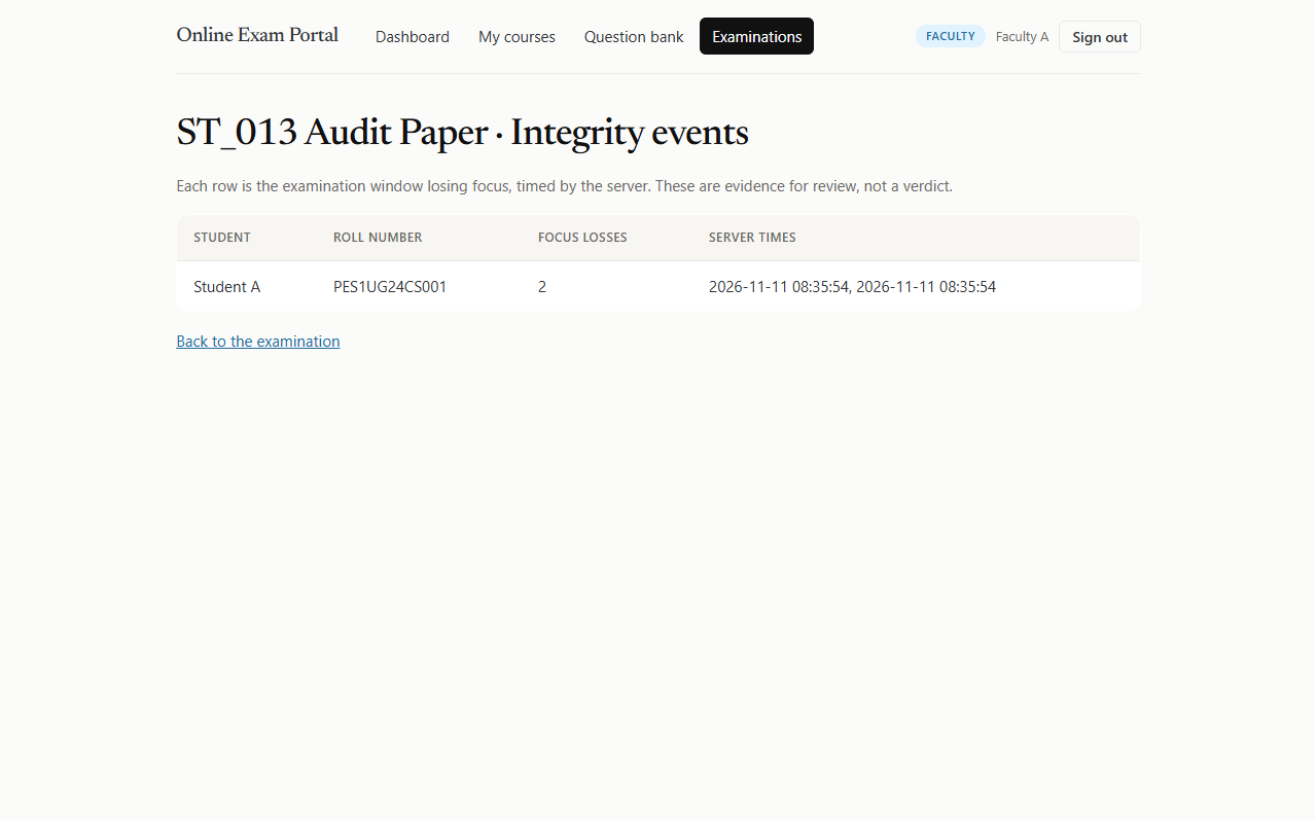
Not proven: neither later run produced a 500, so no traceback was captured. The risk is recorded in Test Plan §10 and Deviations D-18.

5. Defects found

Defect 1 — Moving between questions records false focus-loss events (FR-028)

Severity	**S3 Moderate** (bug triage, Test Plan v5.0 §5.4): the requirement is met only partly and it produces wrong evidence; a human still decides, so no score is affected
Found by	Run 2, 3 Nov; reproduced in Chrome and Firefox
Steps	1: As a Student, start an attempt. 2: Switch away from the tab three times. 3: Move to questions 2, 3 and 4 through the palette, with no further tab switch. 4: Submit. 5: As Faculty, open the exam's integrity page
Expected	3 focus-loss events (FR-028: "switching tabs three times … creates three focus-loss records")
Actual	**7 events**: the 3 real ones plus one for every question page left (FR-028 check). ST_013 shows the same: 6 events for 2 switches
Cause	`attempts/static/attempts/focus.js` reports a loss on `visibilitychange` when the document becomes hidden. The browser also hides the page while navigating away from it, so every move to another question is reported as a focus loss
Fix	11 Nov, after the freeze (Deviations D-15): `focus.js` notes the page's `pagehide` event, which comes before the unload's `visibilitychange`, and reports nothing after it. A tab switch fires no `pagehide`, so it is still reported. `integrity/browser_check_fr028.py` is the browser check: 3 focus losses and 4 question moves give 7 events before the fix and 3 after, in Chrome and Firefox
Re-test	11 Nov, run 3: ST_013 and the FR-028 check re-run through the same script as run 2. ST_013: 2 events for 2 switches, still 2 after answering three questions and submitting; audit log and payloads as in run 2. FR-028: 3 events for 3 switches, 3 after submitting
Status	**Closed** on 11 Nov
Test cases affected	ST_013 (failed in run 2, passes in run 3), FR-028's acceptance criterion; UT_064–UT_067 pass throughout because they test the server side, which works





Why the automated suite missed it. The server records exactly what the page reports, and that is what UT_064–UT_067 test. The extra reports come from the browser script, which the Django test client never runs.

6. Review of v1.0

v1.0 said	v1.1	Why
"No defects found"	Defect 1 (FR-028)	Found by the browser run; v1.0's manual ST_013 looked at the audit log but did not count the focus events
NFR-001, NFR-003: Pass, "Komal's load test, PR #85"	Not verified (v1.1); Pass (v1.3)	PR #85 ran a 5-user smoke test only. v1.1 missed PR #87's 200-user run of 28 Oct; v1.3 adds it and three more runs (§4.1)
NFR-006: Pass	Pass with recorded exceptions	Several PRs were admin-merged without a second reviewer; each is logged in Deviations
NFR-008: consent "refused by the browser's HTML5 required attribute"	Also refused by the server	A browser check alone does not show the server enforces it
NFR-013: CSRF "confirmed by Django middleware (see settings)"	Measured: POST without token → 403	The criterion asks for a refused request, not a setting
ST_008: "covered by ST_009/ST_010 flow tests"	Run directly	ST_008 has its own expected results (refusal before the window, palette states)
Baseline `Documents/…`	`docs/…`	The documents moved to `docs/` for the final demo

7. Limits of this validation

- **Tab switches are simulated.** Playwright's focus emulation means a scripted tab switch fires no focus events, so run 2 fired the `blur` and `focus` events a real switch fires. The page's own script then reported each one. The extra events of Defect 1 come from real navigation, not from the simulation.
- **Fixtures are seeded, not typed.** Users, courses and question banks were created through the ORM, as the Test Plan's preconditions; every step *under test* went through the browser.
- **One machine for the load test.** Locust and Gunicorn shared one laptop, so the client's own time is in every figure. In L1, Gunicorn's own log put the server-side p95 at 240 ms against Locust's 1.9 s (PR #87).

8. Conclusion

On the frozen build 2d74453, **12 of 13 system test cases pass**, and ST_013 fails on Defect 1, a moderate FR-028 defect in the page script that records false focus losses. Defect 1 was fixed on 11 Nov (Deviations D-15), and ST_013 passes on the fixed build, so **13 of 13 pass** there. **Thirteen of 14 non-functional requirements pass**, and NFR-006 passes with exceptions recorded in the Deviations register. NFR-001 and NFR-003 pass the 200-user load test in three of four runs; the fourth, a SQLite lock stall, is disclosed in §4.1. The core examination lifecycle validates end to end in Chrome, Edge and Firefox: set up, attempt with autosave and resume, time-out, automatic evaluation, and publication.